Starts Better and Ends Better: A Target Adaptive Image Signature Tracker

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Abstract

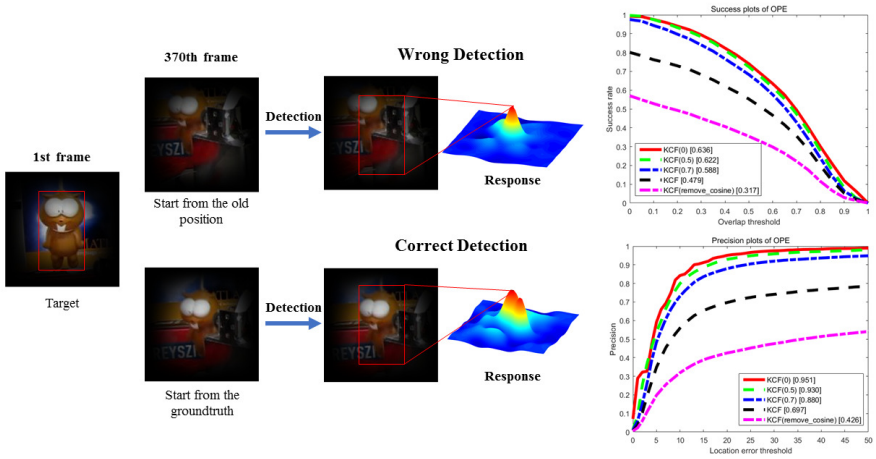
Correlation filter (CF) trackers have achieved outstanding performance in visual object tracking tasks, in which the cosine mask plays an essential role in alleviating boundary effects caused by the circular assumption. However, the cosine mask imposes a larger weight on its center position, which greatly affects CF trackers, that is, their performance will drop significantly if a bad starting point happens to occur. To address the above issue, we propose a target adaptive image signature (TaiS) model to refine the starting point in each frame for CF trackers. Specifically, we incorporate the target priori into the image signature to build a target-specific saliency map, and iteratively refine the starting point with a closed-form solution during the tracking process. As a result, our TaiS is able to find a better starting point close to the center of targets; more importantly, it is independent of specific CF trackers and can efficiently improve their performance. Experiments on two benchmark datasets, i.e., OTB100 and UAV123, demonstrate that our TaiS consistently achieves high performance and updates the state of the arts in visual tracking. [1]

[1] Our code is available at https://github.com/bczhangbczhang/TaiS

1. Introduction

A visual tracker continuously estimates the trajectory of a target given only the initial position and size of the target in the first frame [12, 17, 15]. Correlation filter (CF) trackers have attracted great attention in the visual tracking community, for their unprecedented performance in terms of both speed and accuracy [1, 8, 9, 14, 5, 16]. By involving a group of filters to localize the target in a new frame, the position with maximal correlation response is taken as the new tracked target [6, 18, 3, 5]. To guarantee the efficiency, conventional CF trackers utilize the Fourier transform, which implicitly assumes that the images are periodic. However, the large discontinuity between opposite edges of a non-periodic image will result in a noisy Fourier representation, and thus hurts the tracking performance. To alleviate artifact caused by the discontinuity, correlation filters multiply the images with a cosine mask since 2010 [1], which assigns relatively high weights to the pixels around its center position while abating those far away.

However, such a prevailing scheme of using cosine masks renders the performance of a given tracker to be heavily affected by the starting point, which unfortunately has long been ignored and remains unsolved. We show this by performing a simple experiment with the KCF tracker on the OTB100 dataset. As illustrated in Fig. 1, in each new frame, the starting point P of the detection is set to



(a) (b)

Figure 1. (a) using the traditional starting point that leads to drifting, while a better starting position can make the tracking more robust. (b) the distance precision and the success rate plots of our experiment on examining the impact of the starting point (the dataset is OTB50). α is used to calculate the starting point for each frame.

be $P = \alpha P_{LastFrame} + (1-\alpha)P_{GroundTruth}$, in which $P_{LastFrame}$ [2] is the target position given by the tracker in the last frame and $P_{GroundTruth}$ is the ground-truth position of the target in the new frame. When $\alpha = 0$, KCF starts the detection precisely centering at the target, and the distance precision (threshold = 20) of KCF increases more than 25 %, while the success rate increases by over 20%. Even when $\alpha = 0.7$, the distance precision and the success rate of the original tracker can be improved by a large margin. It is worth mentioning that the cosine mask is indispensable, without which the performance of KCF decreases significantly as shown in Fig. 1.

Inspired by those observations, we propose a target adaptive image signature (TaiS) tracker that incorporates the saliency information to find a better starting point for each frame. Traditionally, the saliency methods are developed to identify the most visually noticeable foreground object efficiently [11, 10, 2]. However, such methods cannot be directly applied to object tracking, especially when multiple objects co-occur in the same scene, because the algorithms mostly fail to take the target to track as the salient objects. We start with a more realistic assumption that the target to track is more salient than the background. Specifically, we leverage the image signature [10] , one of the most important saliency methods that can spatially approximate the sparse foreground of an image, to estimate the target saliency. Moreover, we incorporate the color priori of the target into the image signature based on texture. With the obtained target-adpative saliency map, it enables us to provide a better starting point calculated in a closed form for the CF trackers.

[2] widely used in the existing CF trackers.

To the best of our knowledge, this is the first work that addresses starting points for improving CF trackers. Our method is different from [6], which improves the performance of CF trackers by moderating the boundary effect in training the filters. It is based on the regularization weights that penalize filter values corresponding to features in the background, where the starting point issue remains unsolved. In contrast, the proposed method in this paper provides a general strategy to alleviate the inefficiency caused by the cosine mask which is independent of any specific CF trackers and can be incorporated into other CF trackers.

2. The target adaptive image signature (TaiS) tracker

In this section, we first introduce the basic knowledge of correlation filters, and then analyze the shortcomings of existing correlation filters, especially those caused by the usage of the cosine mask. Finally, a target adaptive image signature is introduced to build a CF tracker.

2.1. The correlation filter trackers

The aim of CF tracker is to learn a multi-channel correlation filter f from a set of training examples $\{(x_k, y_k)\}_1^t$, where $x_k \in \mathbb{R}^d$ is a $d-$dimensional feature map extracted from an image patch of the training sample, y_k is the desired output with a scalar value for the image patch, which is designed to be Gaussian-shaped, and t is the number of training samples. All the training samples are set to be with the same spatial size and extracted from a limited sampling region. We denote the feature layer as x_k^l, $l = \{1, 2, \cdots, d\}$with d as the total number of layers. f^l denotes the correlation filter of the lth feature layer, and the

response of x_k is given by the following equation

$$R_f(x_k) = \sum_{l=1}^{d} x_k^l * f^l, \tag{1}$$

where $*$ denotes the circular convolution, and the filter f is learned by minimizing the optimization problem [6] as

$$f = \arg\min_f \left\{ \sum_{k=1}^{t} \alpha_k ||R_f(x_k) - y_k||^2 + \sum_{l=1}^{d} ||w \odot f^l||^2 \right\}. \tag{2}$$

where w is a regularization weight matrix, and $\odot$ denotes element-wise product.

The starting point affects detection. Most trackers start the target tracking process in a new frame from the position with the maximum response in the last frame. However, this straightforward strategy causes three drawbacks that are not well addressed by the previous work till now:

1. The boundary effect leads to the severe performance degradation in the CF trackers. Based on a mask widely used in existing state-of-the-art CF trackers [6, 7], the weights of the filter are forced to be small at the boundary to suppress the unwanted effect from the boundary. However, they fail to avoid the following two drawbacks.

2. The patches, from both the training samples and the detection candidates, are multiplied by a cosine mask to impose larger weight on the center of the target and also to alleviate the artifact of FFT convolution since 2010 [1]. The cosine mask is indispensable, without which the performance of CF trackers significantly degrade. However, the cosine mask introduces an undesired effect (it pays less attention to the boundary) and causes a tracking failure, very often, when the starting point deviates at the center of target during detection.

3. The errors of CF trackers accumulate. Though theoretically the correct position is obtained based on one of the circulant shifts, a bad starting point for detection will cause severe drifting problem.

Based on facts mentioned above, we propose a target adaptive image signature model that can well approximate the target position in each new frame, which provides a better starting point for CF trackers to improve the tracking performance.

2.2. Target Adaptive Image Signature

The saliency algorithms aim at extracting regions of interest in an image by exploring its fundamental properties. They can be very fast at detecting the foreground objects though they are not precise enough. As discussed before, the performance of CF trackers can be boosted significantly by moving the center of detection region closer to the target. With this intuition in mind, saliency algorithms are perfect tools to be combined with CF trackers. That is to say, the CF trackers are benefited by using the most saliency point as a starting position. However, existing saliency algorithms fail to take advantage of the prior knowledge of the target. Thus, they can not directly be used in the tracking task. In this section, we first introduce image signature, and then propose our target adaptive image signature (TaiS) based on target saliency.

Image Signature: Image signature is a global descriptor of images using Discrete Cosine Transform (DCT) [10]. The image signature can be computed in the frequency domain with low computational cost. For an image I, it is formalized as the following succinct equation to figure out the sparse foreground as

$$ImageSignature(I) = sign(DCT(I)),$$

where $sign(\cdot)$ is the element-wise sign operator, and $DCT(I)$ is the Discrete Cosine Transform of I. We denote the inverse DCT transform as $IDCT(\cdot)$ function, and get the reconstructed image by inversely transforming the image signature back to the spatial domain as $I' = IDCT\{sign(DCT(I))\}$. Then the saliency map S_{IS} is calculated by

$$S_{IS} = G \otimes (I' \odot I'), \tag{3}$$

where $\odot$ is the element-wise product operator, $\otimes$ is the convolution operator, and G is a Gaussian filter. The Gaussian filter is necessary because an object always lies in a contiguous regions rather than discrete pixels.

Target Adaptive Image Signature: Though the image signature is an effective algorithm, it fails to absorb prior knowledge of the target. Hence, it cannot be directly used in object tracking. Here, we introduce our Target Adaptive Image Signature (TaiS) by exploring target prior knowledge; in particular both color and texture are used to calculate the saliency. Our method is enlightened by a color-contrast based saliency [2]. Based on the fact that biological vision system is highly sensitive to contrast, the saliency value is supposed to be higher for a pixel whose color is more distinctive from the others. Obviously such a method can be accelerated by color histogram. Simply, if we quantize the image I into n distinct colors, the saliency value of a color c can be expressed as [2]:

$$S_C(c) = \sum_{i=1}^{n} \rho_i \cdot (D(c, c_i)), \tag{4}$$

where ρ_i denotes the possibility of the ith color c_i of image I, n is the quantization number of color, and $D(\cdot)$ is a distance metric in Lab color space (we use Euclidean distance). The saliency value of a pixel with color c will become larger when c is "far away" from the common colors (with higher

ρ_i). However, such scheme ignores the target information which is indispensable in tracking. Thus, we modify Eq. (4) to be:

$$S'_{Target}(c) = S_T(c)S_C(c). \tag{5}$$

where a greater value of $S_T(c)$ indicates a higher possibility that c belongs to the target. In other words, the pixels with a color close to the target will stand out from all the foreground objects, and $S_T(c)$ is defined as:

$$S_T(c) = \sum_{i=1}^{n^T} w(c_i^T)h(D(c, c_i^T)), \tag{6}$$

where the target I^T is in n^T distinct colors. Let c_i^T be the i th color of the target. $h(\cdot)$ is a monotonically decreasing function, e.g., $h\left(D(c, c_i)\right) = \exp\left(-\frac{D(c,c_i)}{\alpha}\right)$, where α is a positive parameter. $w(c_i^T)$ is the sum of corresponding elements (c_i^T) in M, where M denotes a cosine mask. By doing so, a color c, which is close to c_i^T with large $w(c_i^T)$, will become more salient. Similar to [2], a target adaptive saliency map is obtained via a smoothing operation on Eq. (5) as:

$$S_{Target}(c) = \frac{1}{(m-1)A} \sum_{i=1}^{m} (A - D(c, c_i))S'_{Target}(c_i), \tag{7}$$

where $A = \sum_i D(c, c_i)$, $\frac{1}{(m-1)A}$ is the normalization factor, and c_i is one of the m nearest neighbors of c. However, solely using the color histogram based saliency would not be sufficient to distinguish the target from background due to the similar or same color. To guarantee a better result, S_{IS} (texture) in Eq. (3) and S_{Target} are combined together, which gives a new target adaptive image signature (TaiS) as:

$$S_I = S_{IS} + k \times S_{Target}, \tag{8}$$

where k is a proportional parameter set to be 0.5. Finally, we convolve S_I with the cosine mask M whose size is the same as the target, that is,

$$S = M \otimes S_I. \tag{9}$$

With the saliency map S, the position obtained in Eq. (13) will be taken as the new starting point, which has shown to be a better choice in our extensive experiments. Considering that the color of the target would change during tracking, we adaptively update the weight by:

$$w_{t+1}(c_i^T) = (1-\eta)w_t(c_i^T) + \eta w(c_i^T), \tag{10}$$

where η is a learning rate, and c_i^T is a color belonging to the target.

Algorithm 1 Correlation Filter Tracking with Target Adaptive Image Signature

For a new frame
Input:
Previous target position P_{t-1}.
Detection:
Compute the response map $R_f(z(P_{t-1})) = \mathscr{F}^{-1}\left\{\sum_{l=1}^{d} \hat{z}(P_{t-1})^l \odot \hat{f}^l\right\}$ with the CF tracker. Denote the max response as $\overline{R_f(z(P_{t-1}))}$.
if $\overline{R_f(z(P_{t-1}))} < threshold$ **then**
  Crop a square patch whose center is P_{t-1}.
  Compute the saliency map.
  Compute the new starting point P using Eq. (13).
  Compute the response map $R_f(z(P)) = \mathscr{F}^{-1}\left\{\sum_{l=1}^{d} \hat{z}(P)^l \odot \hat{f}^l\right\}$ with the CF tracker. Denote the max response as $\overline{R_f(z(P))}$.
  New target position $P_t = max(\overline{R_f(z(P))}, \overline{R_f(z(P_{t-1}))})$
else
  $P_t = \overline{R_f(z(P_{t-1}))}$
end if
Training:
Update the target priori and the correlation filter.

2.3. TaiS Correlation Filters

We assist CF trackers with the TaiS model by modifying the detection part with our TaiS. Typically, CF trackers exploit the cyclic shifts of a test sample to efficiently calculate the scores. We denote a candidate patch centered at position P as $z(P)$. By leveraging the convolution property of FFT, we can compute the responses at all locations as

$$R_f(z(P)) = \mathscr{F}^{-1}\left\{\sum_{l=1}^{d} \hat{z}(P)^l \odot \hat{f}^l\right\}, \tag{11}$$

where $\hat{}$ denotes the FFT form of a vector and $\mathscr{F}^{-1}$ denotes the inverse FFT transform. We collect a square patch whose center is the previous position of the target, and then compute the saliency map using the patch. To select the best starting point P, we solve the following optimization problem,

$$P = \arg\min_P \sum_i (||P - P_i||^2 g_i + \lambda_s ||P - P_i||^2 s_i), \tag{12}$$

where P_i is the position of the ith pixel on the saliency map, g_i is a predefined Gaussian-shaped weight whose center is the detection result of the last frame, $s_i \in S$ is calculated based on the saliency map S, and λ_s is the parameter to balance the Gaussian-shaped weight and saliency weight.

We can easily obtain the close-form solution for P as:

$$P = \frac{\sum_i ((g_i + \lambda_s s_i) P_i)}{\sum_i (g_i + \lambda_s s_i)}. \quad (13)$$

Due to the symmetry of the square patch and the Gaussian-shaped weight, our detection method reduces back to the original CF tracker when $\lambda = 0$. Because the computational cost of saliency map is still higher than correlation filters, we only perform the above operations when the response of the original CF tracker is lower than a threshold. In the saliency map, we set the pixels with low saliency value to 0 so that the prediction will not be disturbed by background noise. To increase the stability during tracking, we compare the response of the original detection and modified detection. We pick the result with the larger response as the final result. Algorithm 1 recaptures the whole procedure.

3. Experiments

We validate our TaiS model by integrating it into state-of-the-art trackers, i.e., Efficient Convolution Operater (ECO) [3], Latent Constrained Correlation Filter (LCCF) trackers [18], and CFNet [16]. We conduct extensive experiments on two widely-used benchmarks, i.e., OTB100 [17] and UAV123 [15].

3.1. Implementation Details and Parameters

We apply TaiS on ECO_HC using a combination of HoG and ColorNames features, and ECO_DEEP using the Conv-1 and Conv-5 convolutional layer in VGG-m along with HoG feature, respectively. For LCCF, we also employ both LCCF_HoG and LCCF_DEEP to test our TaiS tracker. LCCF_DEEP exploits the VGG-net-19 as the feature extractor. We use the two-convolutional-layer version of CFNet. We crop a square patch with the size of $4 \times \sqrt{h_T w_T}$ to obtain the saliency map for LCCF and CFNet, where h_T and w_T are the height and width of the target, respectively. It is noted that LCCF is not able to estimate the size of the target, we exploit the method in [4] to calculate Eq. (9).

For fair comparison, we use the similar parameter settings as those in ECO [3], LCCF [18] and CFNet [16]. On UAV123, we set the cell size of HoG in ECO_HC as 4 instead of 6 (the original version) for a better performance on our platform. We set $\lambda_s = 4$ for ECO and $\lambda_s = 5$ for LCCF and CFNet in Eq. (12). The color priori is updated in a sparse fashion as [3], e.g., every 5 frame with a learning rate of $\eta = 0.1$. We adopt a dynamic threshold which is proportional to the average of all the maximum of the past response maps. We note that the saliency map will not be used in the first 4 frames for our trackers.

To trade off between speed and performance, we set different color parameters for different datasets. In the experiments on the OTB100 dataset, the color histogram of the saliency map is divided into 8 bins per color channel with a maximum of distinct colors being 80. For the target patch, we set the maximum of distinct colors to be 30. The UAV123 dataset contains more sequences with background clutters, and we divide the color histogram of the saliency map into 16 bins per color channel.

Our TaiS trackers and their baseline trackers are tested on the same platform with an Intel I7 4.00GHz CPU and a GTX 1080 GPU with 8G RAM for a fair comparison. We also provide the results of [6, 19, 13, 4], which are directly obtained from their official websites or the reference [15].

3.2. Comparison with State-of-the-arts

In this section, we compare our approach with state-of-the-art trackers on two benchmarks. Detailed results of the precision and the overlap success rate are measured and reported with a distance threshold of 20 pixels and the area-under-the-curve (AUC).

The OTB100 Benchmark. The OTB100 dataset is a popular visual tracking dataset containing 100 challenging sequences. In Fig. 2, we compare our approach in terms of precision and success rate with ECO_HC, LCCF_HoG, LCCF_DEEP and CFNet trackers on the OTB100 dataset. We can observe that ECO_HC_TaiS gains the distance precision improvement of 2.4% and the success rate improvement of 1.2% over its baseline ECO_HC, making it able to surpass the state-of-the-art SRDCF_DEEP. LCCF_DEEP_TaiS increases the precision and the success rate by 0.5% and 0.3%, similarly LCCF_HoG_TaiS improves 4.4% and 2.8% over LCCF_HoG. The CFNet_Tais also beats its baseline by a margin of 4.0% and 2.9%. The experiments validate the effectiveness of our tracker when compared with the state-of-the-art CF trackers.

The UAV123 Benchmark. The UAV123 dataset is a also widely-used dataset including 123 sequences for the visual tracking application on unmanned aerial vehicles (UAVs). In comparison with OTB100, UAV123 is more difficult for trackers because the sequences include drastic scale variation, extremely small target, rapid view point change, long-term out-of-view and so on. On UAV123, our TaiS model still achieves better results than the five baseline trackers, including ECO_DEEP, ECO_HC, LCCF_DEEP, LCCF_HoG and CFNet. Fig. 3 illustrates the precision and success rate curves of fourteen trackers. We can observe that in Fig. 3, our TaiS model improve the precision by 0.6%, 2.2% , 5.5%, 5.1% and 2.4% for ECO_DEEP, ECO_HC, LCCF_DEEP , LCCF_HoG and CFNet, and the success rate by 0.5%, 1.7%, 3.1%, 2.7% and 1.7%, respectively. On both datasets, the improvement of LCCF is more phenomenal than the other trackers. It further confirms that our TaiS model is more effective to improve baselines, e.g., LCCF which suffered more from the boundary effect due to the lack of spatial regularization.

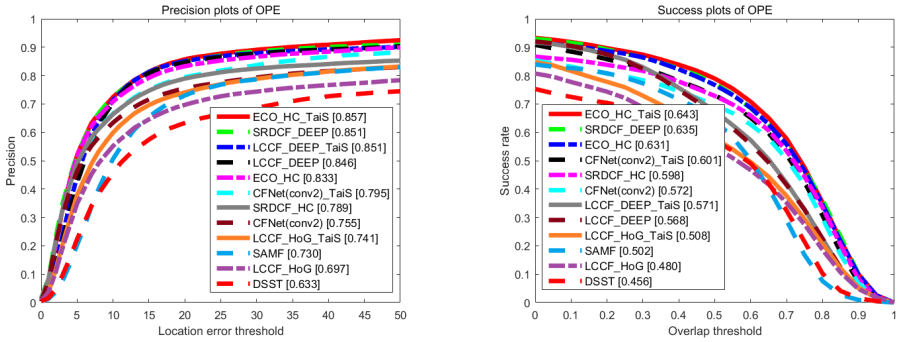


Figure 2. The precision and success rate plots on OTB100 dataset.

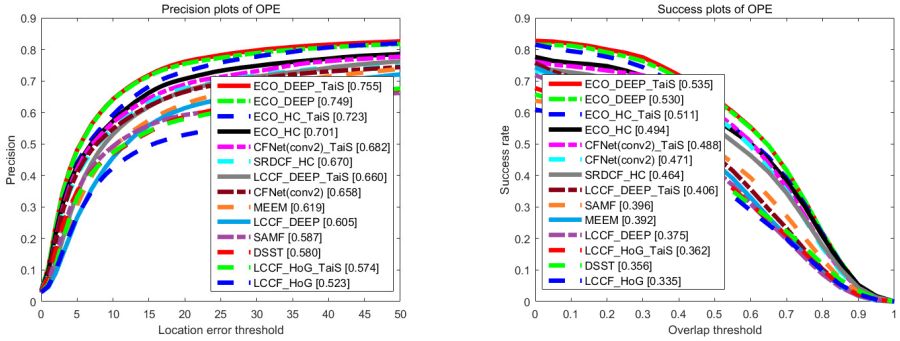


Figure 3. The precision and success rate plots on UAV123 dataset.

3.3. Discussions

In Fig. 4, we compare our TaiS trakers with four baseline trakers in 4 sequences. The saliency maps are also provided. From top to bottom rows, the baseline trackers are respectively ECO_DEEP, ECO_HC, LCCF_HoG and LCCF_DEEP. The first row shows the ability of our approach to recover from long-term out-of-view. Though the target is still in the detection region, ECO_DEEP is not able to recover due to the unsatisfactory starting point. With our TaiS model, the tracker is able to track the target till the end of the sequence. In the second example, ECO_HC fails to track a small target (i.e., a distant truck). Our saliency map is able to detect the small target, provide a better starting point, and thus prevent the tracker from drifting. The third row shows the BlurOwl sequence, where LCCF_HoG faces severe motion blur and a fast-moving target. Because of even heavier boundary artifacts brought by fast motion, the original tracker drifts. By comparison, LCCF_HoG_TaiS is not influenced by the challenging attributes of the sequence. The last row selects the key frames from the wakeboard9 sequence, which shows that the speed and the enormous deformation of the target finally leads the drifting of LCCF_DEEP, while LCCF_DEEP_TaiS successfully tracks the target. Those results have demonstrated the strong ability of our TaiS in challenging tracking tasks.

We also validate the tracking efficiency by comparing TaiS with baselines, ECO_HC and LCCF_HoG. On OTB100, our trackers, both ECO_HC_TaiS and LCCF_HoG_TaiS, still run in real time. On UAV123, our TaiS trackers are running at 18.81 fps and 29.83 fps, which further validate the efficiency of our additional starting point process.

4. Conclusion

We have presented target adaptive image signature (TaiS) for visual tracking. Based on the target adaptive saliency, our TaiS can find a better starting point, which can significantly improve the performance of CF trackers. The TaiS is a highly generalized approach, which is independent of specific CF trackers and can efficiently improve their performance. Experiments on two challenging datasets with five different baseline trackers demonstrate the great effectiveness and high generality of the proposed TaiS for visual tracking.

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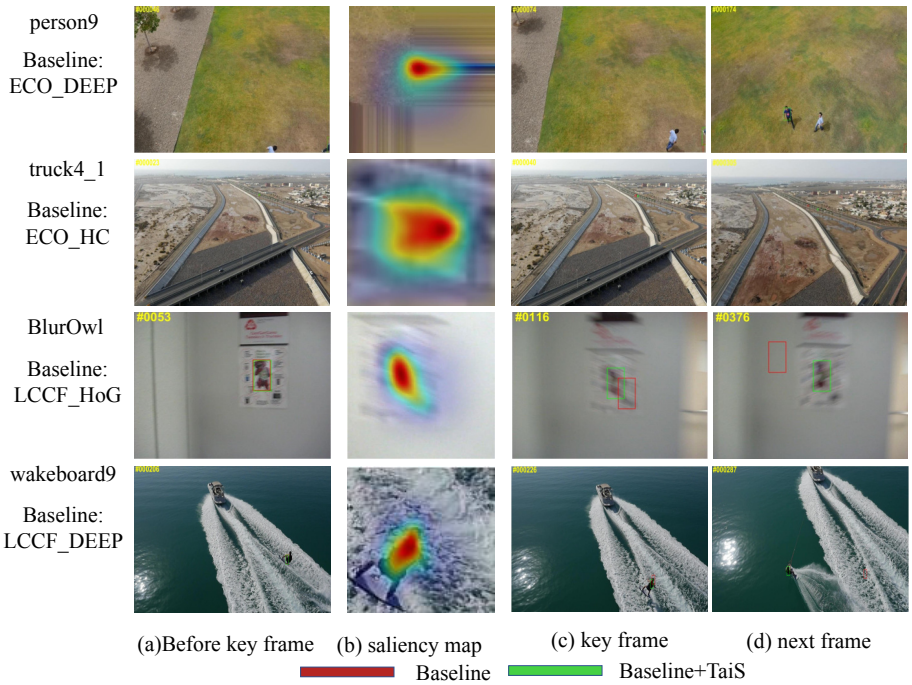

Figure 4. Illustration of some key frames on 4 challenging sequences. The saliency maps in the second column are obtained based on results in the third column. The results show that better starting points lead to better tracking performance.

	ECO_HC	ECO_HC_TaiS	LCCF_HoG	LCCF_HoG_TaiS
OTB100	39.55	28.45	305.9	53.94
UAV123	40.11	18.81	437.57	29.83

Table 1. Speed evaluation of ECO_HC baseline, LCCF_HoG baseline, ECO_HC_TaiS, and LCCF_HoG_TaiS on two datasets.

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